Effective Laplacians, Kron Reduction, and Water-Quality Model Benchmarks

From Electrical Networks to River-Network Regression

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1 Motivation

The full river network contains many interior river nodes at which no water-quality observations are available. Directly assigning a spatial effect to every interior node would greatly increase the size of the Graph-Laplacian regression problem.

Instead, the interior nodes can be eliminated through Kron reduction, producing an effective Laplacian that acts only on the retained water-quality monitoring nodes.

The resulting matrix is

$$\boxed{L_{\text{eff}} = L_{BB} - L_{BI} L_{II}^{-1} L_{IB},}$$

where B denotes retained monitoring nodes and I denotes unobserved interior river nodes.

The reduced Laplacian is not equal to the full Laplacian as a matrix. Rather, it is an exact effective representation of the full graph on the retained nodes.

2 Kron Reduction in Electrical Networks

Kron reduction was originally developed for network reduction in electrical systems. A modern graph-theoretic treatment is given by Dörfler and Bullo [3].

They divide the electrical network into boundary nodes α and interior nodes β and write

$$\begin{bmatrix} I_\alpha \\ I_\beta \end{bmatrix} = \begin{bmatrix} Q_{\alpha\alpha} & Q_{\alpha\beta} \\ Q_{\beta\alpha} & Q_{\beta\beta} \end{bmatrix} \begin{bmatrix} V_\alpha \\ V_\beta \end{bmatrix},$$

where V represents voltage, I represents current injection, and Q is the conductance matrix.

For interior nodes with no external current injection,

$$I_\beta = 0.$$

Therefore,

$$0 = Q_{\beta\alpha}V_\alpha + Q_{\beta\beta}V_\beta,$$

so the interior voltages satisfy

$$\boxed{V_\beta = -Q_{\beta\beta}^{-1}Q_{\beta\alpha}V_\alpha.}$$

Substituting this into the boundary equation gives

$$I_\alpha = Q_{\text{red}}V_\alpha,$$

where

$$\boxed{Q_{\text{red}} = Q_{\alpha\alpha} - Q_{\alpha\beta}Q_{\beta\beta}^{-1}Q_{\beta\alpha}.}$$

Dörfler and Bullo describe this procedure as Gaussian elimination of the interior variables, producing an electrically equivalent reduced network [3].

Thus, the interior nodes are not simply deleted. Their effect on the boundary nodes remains encoded in Q_{red}.

3 Parallel With the River Network

The river-network model uses the same mathematical construction, although the variables have a different interpretation.

The correspondence is

Electrical Network	**River-Network Model**
Q	L
α boundary nodes	B monitoring nodes
β interior nodes	I interior river nodes
V_α	g_B
V_β	g_I
Q_{red}	L_{eff}

Here,

$$g_B$$

represents the spatial graph effects at monitoring locations, while

$$g_I$$

represents the unobserved effects at interior river nodes.

The analogy is mathematical rather than physical. The water-quality graph effects are not voltages or currents; however, both problems involve eliminating unobserved interior graph variables while preserving their effect on retained nodes.

4 Derivation of the Effective Laplacian

Partition the full graph Laplacian as

$$L = \begin{bmatrix} L_{BB} & L_{BI} \\ L_{IB} & L_{II} \end{bmatrix},$$

and partition the graph effect as

$$g = \begin{bmatrix} g_B \\ g_I \end{bmatrix}.$$

For the energy formulation below, L is assumed to be a symmetric, positive-semidefinite weighted graph Laplacian. Furthermore, each connected component must contain at least one retained node, ensuring that L_{II} is invertible.

The Graph-Laplacian energy is

$$g^T L g.$$

Expanding gives

$$g^T L g = g_B^T L_{BB} g_B + 2 g_B^T L_{BI} g_I + g_I^T L_{II} g_I.$$

For fixed monitoring-node effects g_B, the interior values g_I are chosen to minimize this energy.

Differentiating with respect to g_I gives

$$2 L_{IB} g_B + 2 L_{II} g_I = 0.$$

Therefore,

$$\boxed{g_I^* = -L_{II}^{-1} L_{IB} g_B.}$$

Substituting this solution back into the full graph energy gives

$$\boxed{\min_{g_I} \begin{bmatrix} g_B \\ g_I \end{bmatrix}^T L \begin{bmatrix} g_B \\ g_I \end{bmatrix} = g_B^T \left(L_{BB} - L_{BI} L_{II}^{-1} L_{IB} \right) g_B.}$$

Thus,

$$\boxed{L_{\text{eff}} = L_{BB} - L_{BI} L_{II}^{-1} L_{IB}.}$$

The important identity is therefore

$$\boxed{g_B^T L_{\text{eff}} g_B = \min_{g_I} g^T L g.}$$

This shows that the effective Laplacian produces exactly the minimum Graph-Laplacian energy of the full network after the unobserved interior effects have been solved out.

5 Interpretation for the Regression Model

The Graph-Laplacian regression uses a spatial penalty of the form

$$\lambda_L g_B^T L_{\text{eff}} g_B.$$

Using L_{eff} allows the model to estimate graph effects only at the water-quality monitoring locations while still incorporating the influence of the full river network.

Thus,

$$\boxed{\text{full graph + optimal interior effects = effective graph on monitoring nodes.}}$$

This is stronger than simply approximating or deleting the interior river nodes. The Schur complement analytically incorporates their contribution into the reduced matrix.

Dörfler and Bullo also show that effective resistance between retained nodes is preserved under Kron reduction [3]. For retained nodes i and j,

$$\boxed{R_{ij}(L) = R_{ij}(L_{\text{eff}}).}$$

This provides an additional interpretation: the effective connectivity between monitoring locations is preserved even though the interior nodes are no longer represented explicitly.

6 Benchmarking Water-Quality Prediction Models

To compare the proposed graph-based approach with conventional methods, eight models were evaluated across five water-quality parameters: dissolved oxygen (DO) concentration, specific conductance, turbidity, water temperature, and pH.

The comparison includes ordinary least squares (OLS), Ridge regression, Elastic Net, Random Forest, Gradient Boosting, Spline Generalized Additive Models (GAMs), a Dummy baseline, and Graph-Laplacian regression.

All reported results use leave-one-node-out (LOO) prediction. In each outer fold, all observations from one monitoring node are excluded from model fitting and are used only for evaluation. Therefore, the reported R^2 values measure prediction at monitoring locations not used to fit the corresponding model.

Performance was assessed using the coefficient of determination (R^2) and root mean squared error (RMSE), defined by

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2},$$

and

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}.$$

Higher R^2 values and lower RMSE values indicate better predictive performance. RMSE is measured in the units of the corresponding response variable, so its magnitude should not be compared directly across different water-quality parameters.

6.1 Benchmark Results

Table 1 reports the LOO results for the eight models across the five water-quality parameters. The best-performing model for each parameter, as determined by R^2, is highlighted in bold.

Table 1: Leave-one-node-out benchmark results for water-quality prediction. Graph-Laplacian results are included for direct comparison.

Model	Nodes	Rows	R^2	RMSE
Dissolved Oxygen Concentration				
Gradient Boosting	17	114607	**0.735816**	**0.911591**
OLS	17	114607	0.730470	0.920768
Graph-Laplacian	17	114607	0.726100	0.928200

Table 1: Leave-one-node-out benchmark results (continued).

Model	Nodes	Rows	R^2	RMSE
Ridge	17	114607	0.725206	0.929716
Elastic Net	17	114607	0.700178	0.971134
Random Forest	17	114607	0.693250	0.982289
Spline GAM	17	114607	0.645755	1.055600
Dummy	17	114607	-0.028630	1.798774
Specific Conductance				
Graph-Laplacian	24	144980	**0.252500**	**383.553500**
Spline GAM	24	144980	0.030692	436.757198
Gradient Boosting	24	144980	0.001083	443.377755
Dummy	24	144980	-0.062307	457.229359
Random Forest	24	144980	-0.085228	462.135790
Ridge	24	144980	-0.099721	465.211427
Elastic Net	24	144980	-0.116299	468.704787
OLS	24	144980	-0.124152	470.350459
Turbidity				
Gradient Boosting	17	90843	**0.306881**	**469.221329**
Spline GAM	17	90843	0.102329	533.989083
Random Forest	17	90843	0.100274	534.599929
OLS	17	90843	0.047650	550.012078
Ridge	17	90843	0.039319	552.412560
Elastic Net	17	90843	0.036005	553.364354
Graph-Laplacian	17	90843	0.014500	561.813900
Dummy	17	90843	-0.027074	571.182161
Water Temperature				
Ridge	24	144980	**0.863972**	**2.614361**
Elastic Net	24	144980	0.863472	2.619161
OLS	24	144980	0.863421	2.619652
Graph-Laplacian	24	144980	0.862800	2.625900
Spline GAM	24	144980	0.849667	2.748394
Gradient Boosting	24	144980	0.822116	2.989650
Random Forest	24	144980	0.754059	3.515337
Dummy	24	144980	-0.032730	7.203518
pH				
Graph-Laplacian	19	121011	**0.131600**	**0.220800**
Elastic Net	19	121011	0.022639	0.234234
Ridge	19	121011	0.012974	0.235390
OLS	19	121011	0.011565	0.235558
Gradient Boosting	19	121011	0.003533	0.236513
Spline GAM	19	121011	0.002338	0.236655
Random Forest	19	121011	-0.023135	0.239657
Dummy	19	121011	-0.030098	0.240471

6.2 Discussion of Results

The leave-one-node-out comparison shows that the value of graph structure depends strongly on the water-quality parameter.

Water Temperature. Water temperature exhibited the strongest overall predictive performance. Ridge regression achieved the highest R^2 of

$$0.863972.$$

The Graph-Laplacian model was nearly identical at

$$R^2 = 0.862800.$$

This suggests that the seasonal, chemistry, and MERIT predictors already capture most of the predictable variation in water temperature, leaving little additional structure for the graph effect to explain.

Dissolved Oxygen. Gradient Boosting achieved the highest dissolved-oxygen performance with

$$R^2 = 0.735816.$$

The Graph-Laplacian model achieved

$$R^2 = 0.726100,$$

placing it close to OLS and Ridge. Thus, the graph model remains competitive for dissolved oxygen but does not outperform the strongest conventional model.

Turbidity. Gradient Boosting performed substantially better than the other methods, with

$$R^2 = 0.306881.$$

The Graph-Laplacian model achieved only

$$R^2 = 0.014500.$$

This indicates that the variation captured by Gradient Boosting is not well represented by smooth spatial effects over the current river graph. Dynamic nonlinear predictors such as precipitation or runoff may be more important for turbidity.

Specific Conductance. Specific conductance produced the clearest improvement from the Graph-Laplacian model.

The Graph-Laplacian model achieved

$$\boxed{R^2 = 0.252500,}$$

while the best conventional model, Spline GAM, achieved only

$$R^2 = 0.030692.$$

Thus, the graph model explains considerably more leave-one-node-out variation than the conventional regression and machine-learning benchmarks. This provides the strongest evidence in the current results that river-network structure contains useful predictive information for specific conductance.

pH. The Graph-Laplacian model also substantially improved pH prediction.

It achieved

$$\boxed{R^2 = 0.131600,}$$

compared with

$$R^2 = 0.022639$$

for Elastic Net, the strongest conventional benchmark.

Although the absolute R^2 remains modest, the improvement indicates that spatial river-network relationships contribute information that is not captured by the conventional predictor set alone.

6.3 Implications for Graph-Laplacian Regression

The results suggest that the usefulness of the effective Laplacian is target-dependent.

The spatial penalty

$$\boxed{\lambda_L g_B^T L_{\text{eff}} g_B}$$

allows the regression model to use river-network relationships between monitoring locations while still evaluating prediction at completely held-out nodes.

The largest gains occur for specific conductance and pH.

For specific conductance,

$$R^2 : \qquad 0.030692 \longrightarrow 0.252500$$

when comparing the best conventional benchmark with the Graph-Laplacian model.

For pH,

$$R^2 : \qquad 0.022639 \longrightarrow 0.131600.$$

In contrast, the graph model provides little additional benefit for water temperature and dissolved oxygen, where conventional predictors already perform well.

For turbidity, Gradient Boosting performs substantially better than the Graph-Laplacian model.

Therefore, Graph-Laplacian regularization should not be interpreted as a universally superior modeling method. Instead, its benefit appears greatest when the target variable contains spatial structure aligned with the river network that is not already explained by the available covariates.

References

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